XRPL Validator — Daily Engineering Report · 2026-10-19

Period: 2026-10-19 00:00 to 24:00 US Eastern (2026-10-19 04:00 to 2026-10-20 04:00 UTC); ledgers #107,753,848 to #107,775,995. Every count from the ledger and the validator's logs stops at #107,775,995, closed 03:59:52 UTC. Generated 2026-10-21 00:48 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); signing records; Step 1 receipt records (they begin 2026-08-29); Steps 2–5 receipt records (they begin 2026-10-09); deployment records (they begin 2026-09-11); xrpl-validator merge records (they begin 2026-07-04); amendment data; data.xrpl.org agreement scores. Not available: none.

Status

Normal — coverage outside planned maintenance 99.995% (objective 99.9%), every validation agreed with the network, no open engine receipt.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** signed 22,147 of the 22,148 ledgers the network closed (99.995%); every validation matched the network's validated ledger. Unsigned: 0 during planned restarts, 1 outside maintenance (0 incidents, 1 isolated miss).
- **Native engine:** 0 engine receipts recorded.
- **Changes:** 0 deployments, 0 commits merged into main.
- **Network:** 5 slow-close periods; the longest, 02:39 UTC–02:40 UTC, closed ledgers up to 19 s apart.
- **Amendment fixCleanup3_4_0** holds majority; activation due ~2026-10-23 09:34 UTC.

Service levels

Measure	Value
Ledgers closed by the network	22,148
Validations issued	22,147
Coverage (validations / ledgers closed)	99.995%
Validations matching the network's validated ledger hash	22,147 of 22,147

Measure	Value
Unsigned during planned restarts	0
Unsigned outside maintenance	1
Coverage outside planned maintenance (objective 99.9%)	99.995%
Restarts	0
Basis: the hash the trusted quorum named	22,147
External agreement, data.xrpl.org (UTC day 2026-10-19)	0.99995 (1 missed of 19,024)

The basis rows sum to validations issued. Coverage counts the validations the validator issued for ledgers the network closed in the period. Agreement compares each validation's ledger hash with the network's validated hash for that ledger. data.xrpl.org scores UTC days from the validations its crawlers received, so it can differ.

Native engine conformance

Ledgers with a receipt in the period (#107,753,848–#107,775,995), by step:

Step	Ledgers with a receipt
Step 1 · native engine	0
Step 2 · records	0
Step 3 · order	0
Step 4 · peer data	0
Step 4 · validations	0
Step 5 · own hash	0
Writer	0

Live counters are kept only for the current run; none exist for a past day.

Receipt register

No receipts this day.

Incidents

No incidents.

Isolated unsigned ledgers (shorter than 3 ledgers, so not incidents):

- **107,770,853 at 22:20:10 UTC. Cause: The trusted validations for #107,770,853 reached quorum only after those for #107,770,854. Under cycle 183 the validator signed only on a quorum, so it signed #107,770,854 first and could no longer**

sign the older ledger. Cycle 184 signs a ledger as soon as it lands when its own hash matches the reference node's, so a late quorum no longer causes a skip. Fix: commit e6bb9d1 — fix(validator): short of quorum here, sign the hash the reference node validated when our own hash agrees; deployed in cycle 184 at 2026-10-20 13:37 UTC.

Change log

Deployments

No deployments.

Merged into main

No merges into main.

Findings fixed

No engine findings merged into main this day.

Independence program

No independence record for UTC day 2026-10-19; records begin 2026-10-20.

Network

Measure	Value
Ledgers closed	22,148 (#107,753,848–#107,775,995)
Mean close interval	3.90 s
Longest close interval	19 s (before #107,774,768)
Closes at least 15 s apart	5
Empty ledgers	0

- Slow closes 17:43:22 UTC–17:43:40 UTC: ledgers #107,766,625, 1 close at least 15 s apart, up to 18 s.
- Slow closes 17:43:44 UTC–17:44:00 UTC: ledgers #107,766,630, 1 close at least 15 s apart, up to 16 s.
- Slow closes 18:59:52 UTC–19:00:10 UTC: ledgers #107,767,797, 1 close at least 15 s apart, up to 18 s.
- Slow closes 02:39:41 UTC–02:40:00 UTC: ledgers #107,774,768, 1 close at least 15 s apart, up to 19 s.

- Slow closes 02:40:03 UTC–02:40:20 UTC: ledgers #107,774,772, 1 close at least 15 s apart, up to 17 s.

From ledger close times, which the network records at a 10-second resolution or coarser, so single intervals are approximate.

Amendment watch

Read from the XRP Ledger: the Amendments object as of the last ledger of the day, and the day's EnableAmendment pseudo-transactions.

Amendment	Majority since	Activation due
fixCleanup3_4_0	2026-10-09 09:34 UTC	2026-10-23 09:34 UTC

Amendments enabled on mainnet: 97.

Open items

- Amendment fixCleanup3_4_0: activation due ~2026-10-23 09:34 UTC.

Corrections to earlier reports

None.